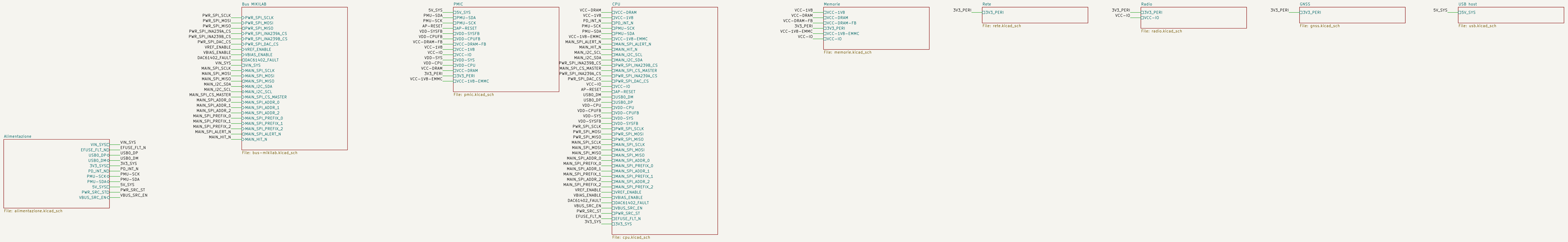
Alimentazione
VIN_SYS
EFUSE_FLT_N
USB0_DP
USB0_DM
3V3_SYS
PD_INT_N
PMU-SCK
PMU-SDA
5V_SYS
PWR_SRC_ST
VBUS_SRC_EN
File: alimentazione.kicad_sch
Bus MIKILAB
PWR_SPI_SCLK
PWR_SPI_MOSI
PWR_SPI_MISO
PWR_SPI_INA239A_CS
PWR_SPI_INA239B_CS
PWR_SPI_DAC_CS
VREF_ENABLE
VBIAS_ENABLE
DAC61402_FAULT
VIN_SYS
MAIN_SPI_SCLK
MAIN_SPI_MOSI
MAIN_SPI_MISO
MAIN_I2C_SDA
MAIN_I2C_SCL
MAIN_SPI_CS_MASTER
MAIN_SPI_ADDR_0
MAIN_SPI_ADDR_1
MAIN_SPI_ADDR_2
MAIN_SPI_PREFIX_0
MAIN_SPI_PREFIX_1
MAIN_SPI_PREFIX_2
MAIN_SPI_ALERT_N
MAIN_HIT_N
File: bus-mikilab.kicad_sch
PMIC
5V_SYS
PMU-SDA
PMU-SCK
AP-RESET
VDD-SYSFB
VDD-CPUFB
VCC-DRAM-FB
VCC-1V8
VCC-IO
VDD-SYS
VDD-CPU
VCC-DRAM
3V3_PERI
VCC-1V8-EMMC
File: pmic.kicad_sch
CPU
VCC-DRAM
VCC-1V8
PD_INT_N
PMU-SCK
PMU-SDA
VCC-1V8-EMMC
MAIN_SPI_ALERT_N
MAIN_HIT_N
MAIN_I2C_SCL
MAIN_I2C_SDA
PWR_SPI_INA239B_CS
MAIN_SPI_CS_MASTER
PWR_SPI_INA239A_CS
PWR_SPI_DAC_CS
VCC-IO
AP-RESET
USB0_DM
USB0_DP
VDD-CPU
VDD-CPUFB
VDD-SYS
VDD-SYSFB
PWR_SPI_SCLK
PWR_SPI_MOSI
PWR_SPI_MISO
MAIN_SPI_SCLK
MAIN_SPI_MOSI
MAIN_SPI_MISO
MAIN_SPI_ADDR_0
MAIN_SPI_PREFIX_0
MAIN_SPI_ADDR_1
MAIN_SPI_PREFIX_1
MAIN_SPI_ADDR_2
MAIN_SPI_PREFIX_2
VREF_ENABLE
VBIAS_ENABLE
DAC61402_FAULT
VBUS_SRC_EN
PWR_SRC_ST
EFUSE_FLT_N
3V3_SYS
File: cpu.kicad_sch
Memorie
VCC-1V8
VCC-DRAM
VCC-DRAM-FB
3V3_PERI
VCC-1V8-EMMC
VCC-IO
File: memorie.kicad_sch
Rete
3V3_PERI
File: rete.kicad_sch
Radio
3V3_PERI
VCC-IO
File: radio.kicad_sch
GNSS
3V3_PERI
File: gnss.kicad_sch
USB host
5V_SYS
File: usb.kicad_sch

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CPU: Allwinner H616 (TFBGA-284, 0.65 mm). Ball map solo dal datasheet H616 V1.0 (tabella 4-2, verificata sulla figura 7-1). Serie 3xx.

AC_VRA1 su C17 (figura 7-1, footprint EasyEDA e schema Orange Pi Zero2 v1.3; la tabella 4-2 dice D17 per errore). Mappa completa dei pin in docs/pinmap-h616.md.

Net DRAM_*/EMMC_*/SD_*/WL_*/BT_*/RGMII_*/PHY_*/GNSS_*/WIFI_*/USB1_*/SPI1_* globali verso gli altri fogli.

Clock: quarzo 24 MHz su DXIN/DXOUT (18 pF, 0R in serie su DXOUT), DXLDO_OUT 220 nF. L'H616 non ha pin per il quarzo 32 kHz: X32KFOUT (PG10) e' derivato internamente.

RESET (dominio VCC_PLL 1.8 V) dal PWROK dell'AXP313A (10k a VCC-1V8 sul foglio PMIC, 1 nF qui) + tasto; FEL a GND con tasto (10k a VCC-IO). JTAG_SEL NC come sulla Zero2 (JTAG su PF). PLLTEST, LRADC liberi.

J301: 1 = TX del SoC (PH0), 2 = RX del SoC (PH1), 3 = GND. Livelli 3.3 V.

Disaccoppiamento vicino alle sfere. 100 nF su VCC-PLL, VCC_DCXO, VDD18_DRAM, VCC_HDMI, VCC_TV (VCC-1V8) e su VCC_IO, VCC_PG/PI, VCC_USB, VCC_EPHY (VCC-IO), come la Zero2.

Analogica (codec e AC300 non usati): VRA1/VRA2 470 nF verso AGND, AGND a GND con 0R, AC_VRA1 1 uF. ZQ del controller DRAM 240R 1% a GND.

Come Orange Pi Zero2 (H616) v1.3: REXT (K18) 150k 1% ad AGND (R337), AVCC 2.2 uF (C334), AC_AVCC 2.2 uF (C335), AC_VRA1 1 uF su C17.

SPI1 (PH6 CLK, PH7 MOSI, PH8 MISO) condiviso tra BUS_POWER e MAIN, chip select con cs-gpios. 33R serie per ramo su CLK/MOSI/MISO; EEPROM MAC locale su SPI1_* (foglio Rete).

Le MISO dei due rami si uniscono dopo i 33R: tutti gli slave (INA239, DAC61402, bus MAIN) devono tenere SDO in alta impedenza con CS alto. ADDR/PREFIX: 100R serie.

Livelli di riposo: CS e CS_MASTER alti (10k), VREF_ENABLE basso (10k a GND), VBIAS_ENABLE basso (100k a GND), ALERT# e HIT# 4.7k, I2C del bus 2.2k, FAULT 10k.

ESD bassa capacita TPD4E05U06DQA (0.5 pF, 5.5 V) sulle linee logiche di J1/J2, vicino ai connettori. Nessuna protezione su V_BIAS e riferimenti.

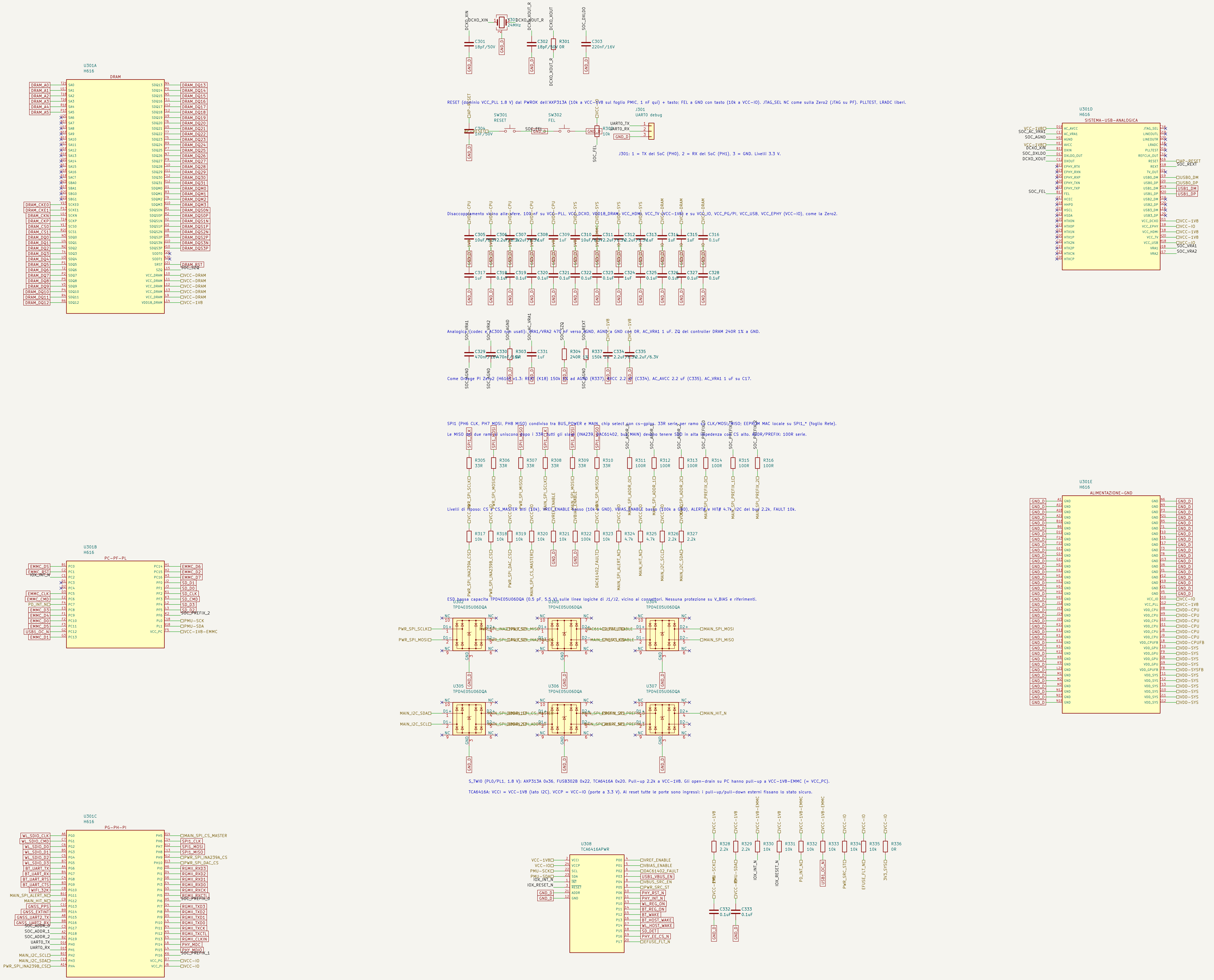
S_TWI0 (PL0/PL1, 1.8 V): AXP313A 0x36, FUSB302B 0x22, TCA6416A 0x20. Pull-up 2.2k a VCC-1V8. Gli open-drain su PC hanno pull-up a VCC-1V8-EMMC (= VCC_PC).

TCA6416A: VCCI = VCC-1V8 (lato I2C), VCCP = VCC-IO (porte a 3.3 V). Al reset tutte le porte sono ingressi: i pull-up/pull-down esterni fissano lo stato sicuro.

Porte: P00 VREF_ENABLE, P01 VBIAS_ENABLE, P02 DAC61402_FAULT (in), P03 USB1_VBUS_EN, P04 VBUS_SRC_EN, P05 PWR_SRC_ST (in), P06 PHY_RST_N, P07 PHY_INT_N (in),

P10 WL_REG_ON, P11 BT_REG_ON, P12 BT_WAKE, P13 BT_HOST_WAKE (in), P14 WL_HOST_WAKE (in), P15 SD_DET (in), P16 PHY_EE_CS_N, P17 EFUSE_FLT_N (in).

R336: 3V3_SYS (VDD del FUSB302B) da VCC-IO.



U301A
H616
DRAM
U301B
H616
PC-PF-PL
U301C
H616
PG-PH-PI
U3010
H616
SISTEMA-USB-ANALOGICA
U301E
H616
ALIMENTAZIONE-GND
Y301
24MHz
C301
18pF/50V
C302
18pF/50V
R301
0R
C303
220nF/16V
DCXO_XIN
DCXO_XOUT_R
DCXO_XOUT
SOC_DXLDO
C304
1nF/50V
SW301
RESET
SW302
FEL
R302
10k
J301
UART0 debug
UART0_TX
UART0_RX
SOC_FEL
AP-RESET
U302
TPD4E05U06DQA
U303
TPD4E05U06DQA
U304
TPD4E05U06DQA
U305
TPD4E05U06DQA
U306
TPD4E05U06DQA
U307
TPD4E05U06DQA
U308
TCA6416APWR
GND_D

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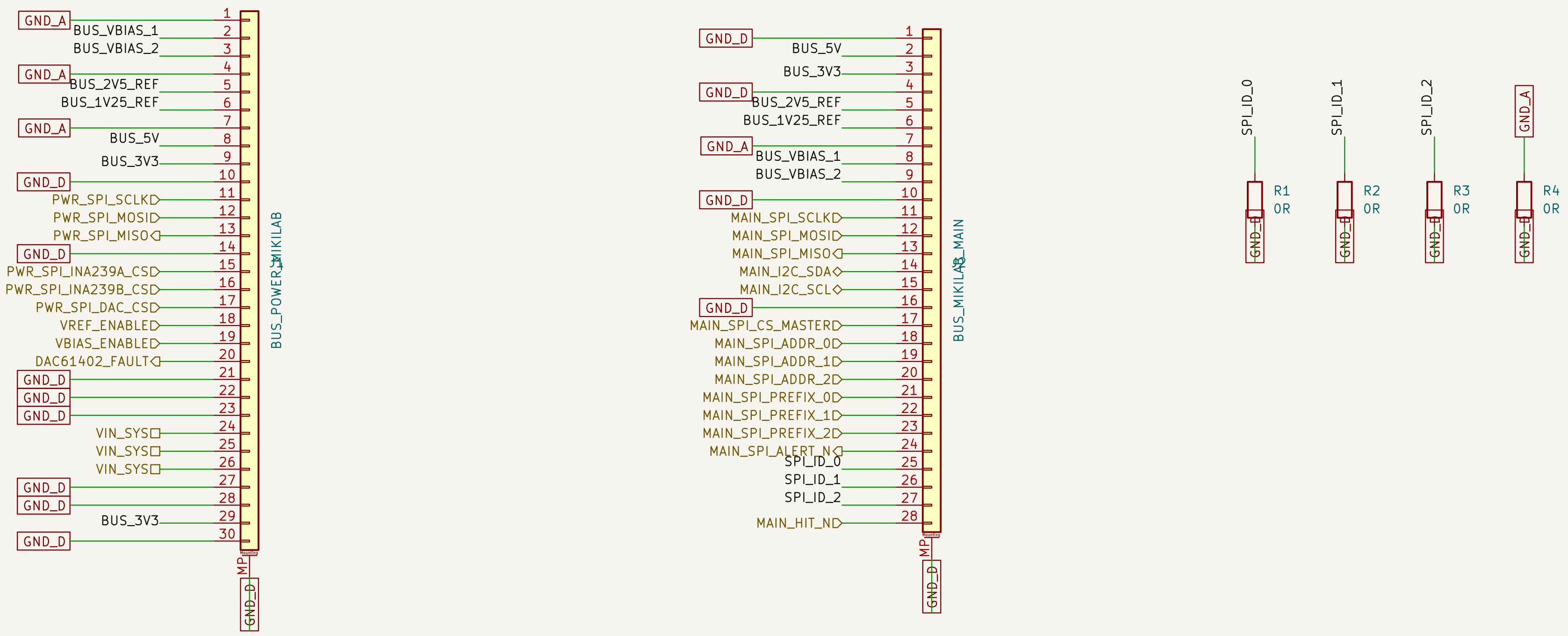
J1 BUS_POWER 30 pin (Hirose FH12-30S): pin 1-22 come J3 del modulo Power, 24-26 VIN_SYS 12 V, 29 secondo 3V3. Rail BUS_* passanti verso J2, non collegati ai rail della scheda.

VIN_SYS -> eFuse TPS259470 -> buck TPS56637 12->5 V (foglio Alimentazione). BUS_5V va solo a J2 pin 2.

J2 MAIN 28 pin secondo BPS-MIKILAB-2026-REV5.1 (pin 28 = HIT#) (spec: XUNPU FPC-05F-28PH20; footprint H616:FPC-05F-28PH20 da EasyEDA C2856807).

Sul foglio CPU: SPI1 condiviso BUS_POWER/MAIN con 33R serie per ramo, 100R su ADDR/PREFIX, pull-up 4.7k su ALERT#/HIT#, 2.2k su I2C, 10k sui CS, ESD TPD4E05U06.

R4 0R: ponte GND_A-GND_D, DNP di default (sul modulo Power GND_A e GND_D sono gia la stessa net). R1-R3: ID iniziale 000 (scheda = J_OUT, testa della catena).


J1
BUS_POWER_MIKILAB
GND_A 1
BUS_VBIAS_1 2
BUS_VBIAS_2 3
GND_A 4
BUS_2V5_REF 5
BUS_1V25_REF 6
GND_A 7
BUS_5V 8
BUS_3V3 9
GND_D 10
PWR_SPI_SCLK 11
PWR_SPI_MOSI 12
PWR_SPI_MISO 13
GND_D 14
PWR_SPI_INA239A_CS 15
PWR_SPI_INA239B_CS 16
PWR_SPI_DAC_CS 17
VREF_ENABLE 18
VBIAS_ENABLE 19
DAC61402_FAULT 20
GND_D 21
GND_D 22
GND_D 23
VIN_SYS 24
VIN_SYS 25
VIN_SYS 26
GND_D 27
GND_D 28
BUS_3V3 29
GND_D 30
MP
GND_D
J2
BUS_MIKILAB_MAIN
GND_D 1
BUS_5V 2
BUS_3V3 3
GND_D 4
BUS_2V5_REF 5
BUS_1V25_REF 6
GND_A 7
BUS_VBIAS_1 8
BUS_VBIAS_2 9
GND_D 10
MAIN_SPI_SCLK 11
MAIN_SPI_MOSI 12
MAIN_SPI_MISO 13
MAIN_I2C_SDA 14
MAIN_I2C_SCL 15
GND_D 16
MAIN_SPI_CS_MASTER 17
MAIN_SPI_ADDR_0 18
MAIN_SPI_ADDR_1 19
MAIN_SPI_ADDR_2 20
MAIN_SPI_PREFIX_0 21
MAIN_SPI_PREFIX_1 22
MAIN_SPI_PREFIX_2 23
MAIN_SPI_ALERT_N 24
SPI_ID_0 25
SPI_ID_1 26
SPI_ID_2 27
MAIN_HIT_N 28
MP
GND_D
SPI_ID_0
R1
0R
GND_D
SPI_ID_1
R2
0R
GND_D
SPI_ID_2
R3
0R
GND_D
GND_A
R4
0R
GND_D

Sheet: /Bus MIKILAB/
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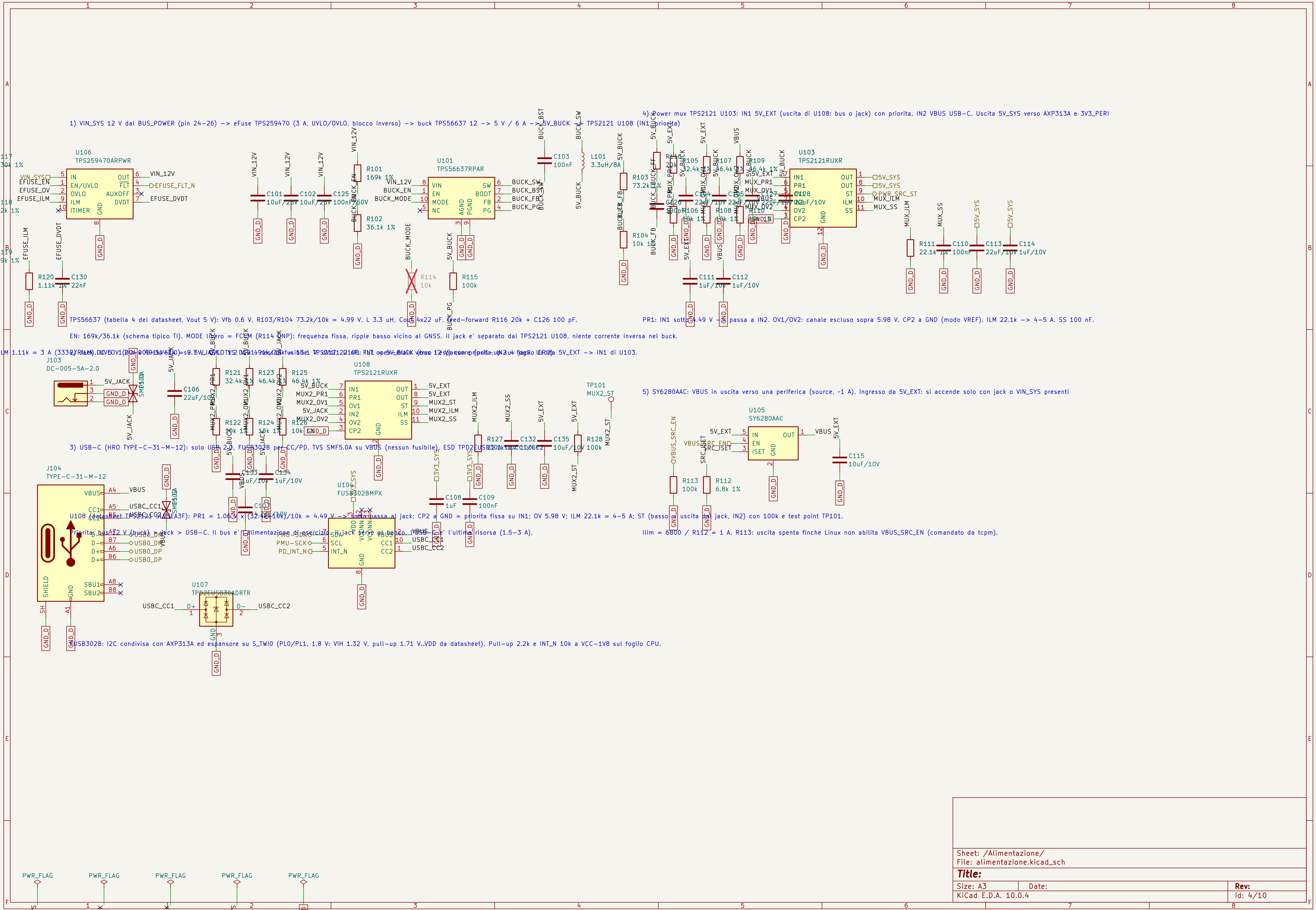

1) VIN_SYS 12 V dal BUS_POWER (pin 24–26) –> eFuse TPS259470 (3 A, UVLO/OVLO, blocco inverso) –> buck TPS56637 12 –> 5 V / 6 A –> 5V_BUCK –> TPS2121 U108 (IN1 priorita)

4) Power mux TPS2121 U103: IN1 5V_EXT (uscita di U108: bus o jack) con priorita, IN2 VBUS USB–C. Uscita 5V_SYS verso AXP313A e 3V3_PERI

TPS56637 (tabella 4 del datasheet, Vout 5 V): Vfb 0.6 V, R103/R104 73.2k/10k = 4.99 V, L 3.3 uH, Cout 4x22 uF, feed–forward R116 20k + C126 100 pF.

EN: 169k/36.1k (schema tipico TI). MODE libero = FCCM (R114 DNP): frequenza fissa, ripple basso vicino al GNSS. Il jack e' separato dal TPS2121 U108, niente corrente inversa nel buck.

eFuse: RILM 1.11k = 3 A (3334/RILM). [illegible] 5V_EXT –> IN1 di U103.

PR1: IN1 sotto 4.49 V –> passa a IN2. OV1/OV2: canale escluso sopra 5.98 V. CP2 a GND (modo VREF). ILM 22.1k –> 4–5 A. SS 100 nF.

5) SY6280AAC: VBUS in uscita verso una periferica (source, ~1 A). Ingresso da 5V_EXT: si accende solo con jack o VIN_SYS presenti

3) USB–C (HRO TYPE–C–31–M–12): solo USB 2.0, FUSB302B per CC/PD, TVS SMF5.0A su VBUS (nessun fusibile), ESD TPD2EUSB30A su CC1/CC2

U108 (datasheet TPS2121 SLVSEA3F): PR1 = 1.06 V x (32.4k+10k)/10k = 4.49 V –> sotto passa al jack; CP2 a GND = priorita fissa su IN1; OV 5.98 V; ILM 22.1k = 4–5 A; ST (basso = uscita dal jack, IN2) con 100k e test point TP101.

Priorita: bus 12 V (buck) > jack > USB–C. Il bus e' l'alimentazione di esercizio, il jack serve al banco, l'USB–C e' l'ultima risorsa (1.5–3 A).

Ilim = 6800 / R112 = 1 A. R113: uscita spenta finche Linux non abilita VBUS_SRC_EN (comandato da tcpm).

FUSB302B: I2C condivisa con AXP313A ed espansore su S_TWI0 (PL0/PL1, 1.8 V: VIH 1.32 V, pull–up 1.71 V..VDD da datasheet). Pull–up 2.2k e INT_N 10k a VCC–1V8 sul foglio CPU.

U106
TPS259470ARPWR
U101
TPS56637RPAR
L101
3.3uH/8A
U103
TPS2121RUXR
J103
DC–005–5A–2.0
U108
TPS2121RUXR
TP101
MUX2_ST
U105
SY6280AAC
J104
TYPE–C–31–M–12
U104
FUSB302BMPX
U107
TPD2EUSB30ADRTR
PWR_FLAG
VIN_SYS
5V_BUCK
5V_JACK
3V3_SYS
GND_D

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PMIC AXP313A (datasheet X-Powers v1.0) come su Orange Pi Zero3 (H618 + AXP313A): VIN 3.0–5.5 V da 5V_SYS, I2C 0x36 su S_TWI0 (PL0/PL1). Serie 2xx.

DCDC1 VDD-SYS 0.9 V (VDD_SYS + VDD_GPU, max 0.99 V) 1.5 A, DCDC2 VDD-CPU 0.81–1.1 V 3 A, DCDC3 VCC-DRAM 1.1 V (LPDDR4 VDD2/VDDQ) 2 A,

ALDO1 VCC-1V8 300 mA (VCC_PLL, VDD18_DRAM, DCXO, AVCC, AC_AVCC, HDMI, TV, LPDDR4 VDD1, TCA6416A VCCI), DLDO1 VCC-IO 3.3 V 500 mA (VCC_IO/PG/PI/USB). RTCLDO non usato (H616 senza VCC_RTC).

Ingressi VIN1/VIN2/VIN3: 10 uF + 100 nF ciascuno vicino ai pin. VREF 1 uF (datasheet).

DCDC: induttori 1 uH (datasheet: Isat > 1.3 x carico, DCR < 30 mohm), uscite >= 10 uF. FB con sense remoto: DCDC1 da VDD_GPUFB (F8), DCDC2 da VDD_CPUFB (L8), DCDC3 dalla LPDDR4.

LDO: ALDO1 VCC-1V8 e DLDO1 VCC-IO con 4.7 uF (datasheet: >= 1 uF). RTCLDO libero (LDO non usato: pin flottante, nessun condensatore).

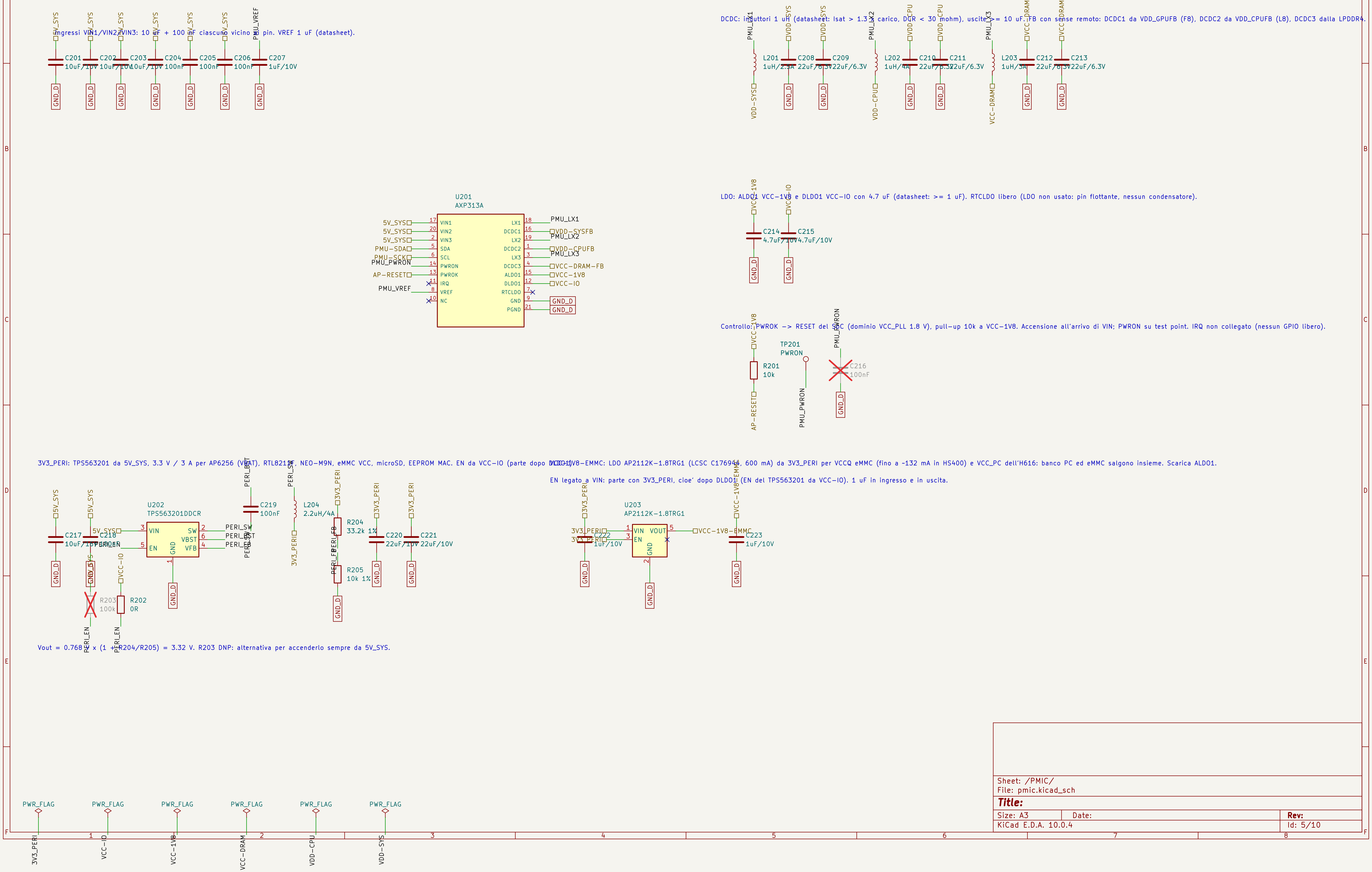
Controllo: PWROK -> RESET del SoC (dominio VCC_PLL 1.8 V), pull-up 10k a VCC-1V8. Accensione all'arrivo di VIN; PWRON su test point. IRQ non collegato (nessun GPIO libero).

3V3_PERI: TPS563201 da 5V_SYS, 3.3 V / 3 A per AP6256 (VBAT), RTL8211F, NEO-M9N, eMMC VCC, microSD, EEPROM MAC. EN da VCC-IO (parte dopo DLDO1).

VCC-1V8-EMMC: LDO AP2112K-1.8TRG1 (LCSC C176944, 600 mA) da 3V3_PERI per VCCQ eMMC (fino a ~132 mA in HS400) e VCC_PC dell'H616: banco PC ed eMMC salgono insieme. Scarica ALDO1.

EN legato a VIN: parte con 3V3_PERI, cioe' dopo DLDO1 (EN del TPS563201 da VCC-IO). 1 uF in ingresso e in uscita.

Vout = 0.768 V x (1 + R204/R205) = 3.32 V. R203 DNP: alternativa per accenderlo sempre da 5V_SYS.



5V_SYS
PMU_VREF
C201 10uF/10V
C202 10uF/10V
C203 10uF/10V
C204 100nF
C205 100nF
C206 100nF
C207 1uF/10V
GND_D
PMU_LX1
VDD-SYS
L201 1uH/2.5A
C208 22uF/6.3V
C209 22uF/6.3V
PMU_LX2
VDD-CPU
L202 1uH/4A
C210 22uF/6.3V
C211 22uF/6.3V
PMU_LX3
VCC-DRAM
L203 1uH/3A
C212 22uF/6.3V
C213 22uF/6.3V
U201
AXP313A
VIN1 17
VIN2 20
VIN3 2
SDA 5
SCL 6
PWRON 14
PWROK 13
IRQ 11
VREF 8
NC 10
LX1 18
DCDC1 16
LX2 19
DCDC2 1
LX3 3
DCDC3 4
ALDO1 15
DLDO1 12
RTCLDO 7
GND 9
PGND 21
PMU-SDA
PMU-SCK
PMU_PWRON
AP-RESET
VDD-SYSFB
VDD-CPUFB
VCC-DRAM-FB
VCC-1V8
VCC-IO
C214 4.7uF/10V
C215 4.7uF/10V
TP201 PWRON
R201 10k
PMU_PWRON
C216 100nF
U202
TPS563201DDCR
VIN 3
EN 5
SW 2
VBST 6
VFB 4
GND 1
PERI_SW
PERI_BST
PERI_FB
PERI_EN
C217 10uF/10V
C218 100nF
C219 100nF
L204 2.2uH/4A
R204 33.2k 1%
R205 10k 1%
3V3_PERI
C220 22uF/10V
C221 22uF/10V
R203 100k
R202 0R
U203
AP2112K-1.8TRG1
VIN 1
EN 3
VOUT 5
GND 2
C222 1uF/10V
C223 1uF/10V
VCC-1V8-EMMC
PWR_FLAG
3V3_PERI
VCC-IO
VCC-1V8
VCC-DRAM
VDD-CPU
VDD-SYS

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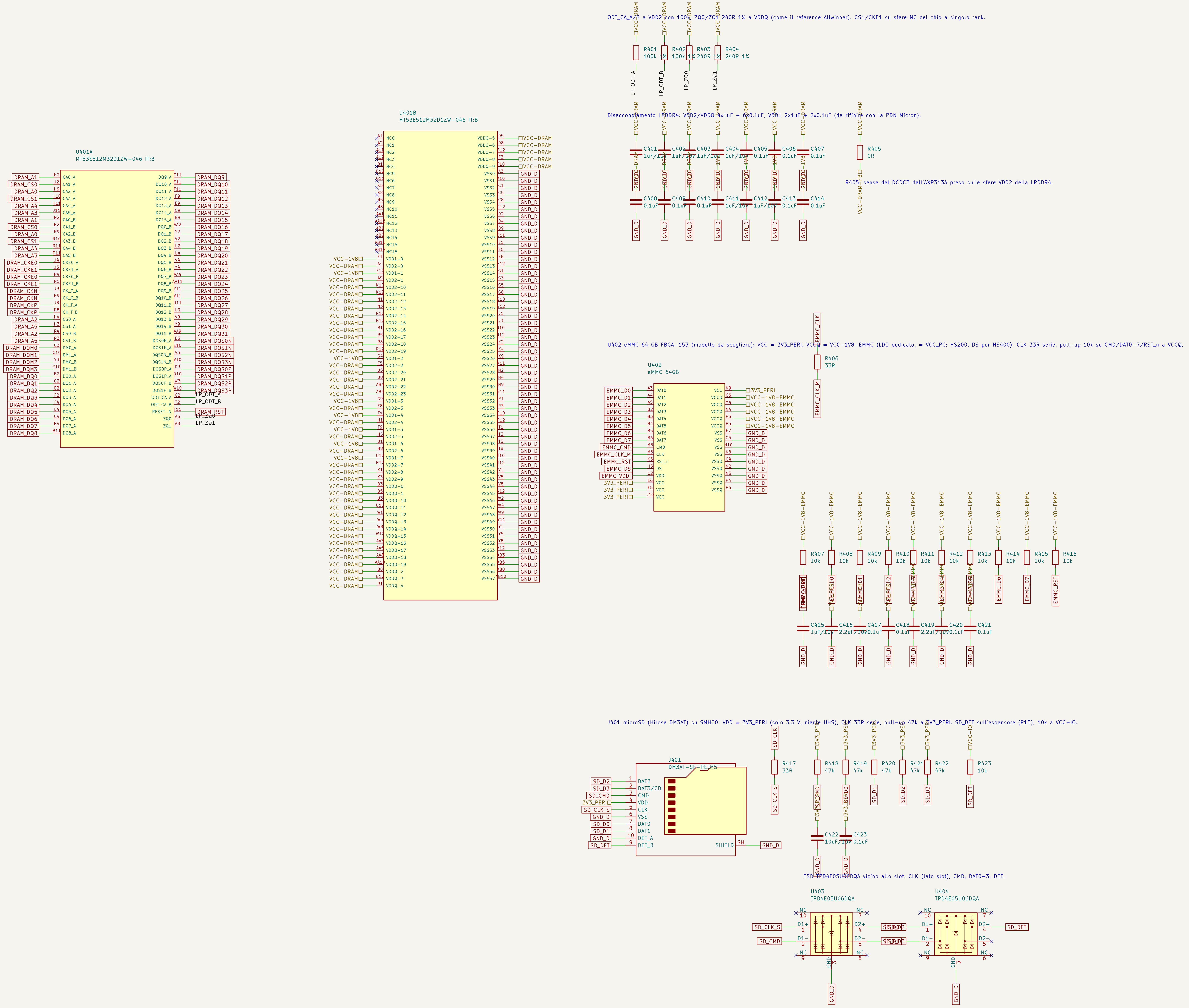
MEMORIE: LPDDR4 2 GB x32 (1 chip, come Orange Pi Zero3), eMMC 8 bit su SMHC2 (PC, 1.8 V), microSD su SMHC0 (PF, 3.3 V). Serie 4xx.

U401 MT53E512M32D1ZW-046 IT:B: 1 die, 2 canali x16, CA comuni ai due canali. VDD1 = VCC-1V8 (ALDO1), VDD2/VDDQ = VCC-DRAM 1.1 V (DCDC3). Niente VTT/VREF/VPP.

CA0 <- SA1, CA1 <- SCS0, CA2 <- SA0, CA3 <- SCS1, CA4 <- SA4, CA5 <- SA3, CS0 <- SA2, CS1 <- SA5 (da phy_init U-Boot, NON VERIFICATO). DQ dritti: canale A = byte 0-1, canale B = byte 2-3.


U401A
MT53E512M32D1ZW-046 IT:B
U401B
MT53E512M32D1ZW-046 IT:B
R401 100k 1%
R402 100k 1%
R403 240R 1%
R404 240R 1%
LP_ODT_A
LP_ODT_B
LP_ZQ0
LP_ZQ1
C401 1uF/10V
C402 1uF/10V
C403 1uF/10V
C404 1uF/10V
C405 0.1uF
C406 0.1uF
C407 0.1uF
C408 0.1uF
C409 0.1uF
C410 0.1uF
C411 1uF/10V
C412 1uF/10V
C413 0.1uF
C414 0.1uF
R405 0R
VCC-DRAM
VCC-1V8
GND_D
U402
eMMC 64GB
R406 33R
EMMC_CLK
EMMC_CLK_M
3V3_PERI
VCC-1V8-EMMC
R407 10k
R408 10k
R409 10k
R410 10k
R411 10k
R412 10k
R413 10k
R414 10k
R415 10k
R416 10k
C415 1uF/10V
C416 2.2uF/10V
C417 0.1uF
C418 0.1uF
C419 2.2uF/10V
C420 0.1uF
C421 0.1uF
J401
DM3AT-SF-PEJM5
DAT2
DAT3/CD
CMD
VDD
CLK
VSS
DAT0
DAT1
DET_A
DET_B
SHIELD
R417 33R
R418 47k
R419 47k
R420 47k
R421 47k
R422 47k
R423 10k
C422 10uF/10V
C423 0.1uF
SD_CLK
SD_CLK_S
SD_CMD
SD_D0
SD_D1
SD_D2
SD_D3
SD_DET
VCC-IO
U403
TPD4E05U06DQA
U404
TPD4E05U06DQA

ODT_CA_A/B a VDD2 con 100k, ZQ0/ZQ1 240R 1% a VDDQ (come il reference Allwinner). CS1/CKE1 su sfere NC del chip a singolo rank.

Disaccoppiamento LPDDR4: VDD2/VDDQ 4x1uF + 6x0.1uF, VDD1 2x1uF + 2x0.1uF (da rifinire con la PDN Micron).

R405: sense del DCDC3 dell'AXP313A preso sulle sfere VDD2 della LPDDR4.

U402 eMMC 64 GB FBGA-153 (modello da scegliere): VCC = 3V3_PERI, VCCQ = VCC-1V8-EMMC (LDO dedicato, = VCC_PC: HS200, DS per HS400). CLK 33R serie, pull-up 10k su CMD/DAT0-7/RST_n a VCCQ.

J401 microSD (Hirose DM3AT) su SMHC0: VDD = 3V3_PERI (solo 3.3 V, niente UHS), CLK 33R serie, pull-up 47k a 3V3_PERI. SD_DET sull'espansore (P15), 10k a VCC-IO.

ESD TPD4E05U06DQA vicino allo slot: CLK (lato slot), CMD, DAT0-3, DET.

Sheet: /Memorie/
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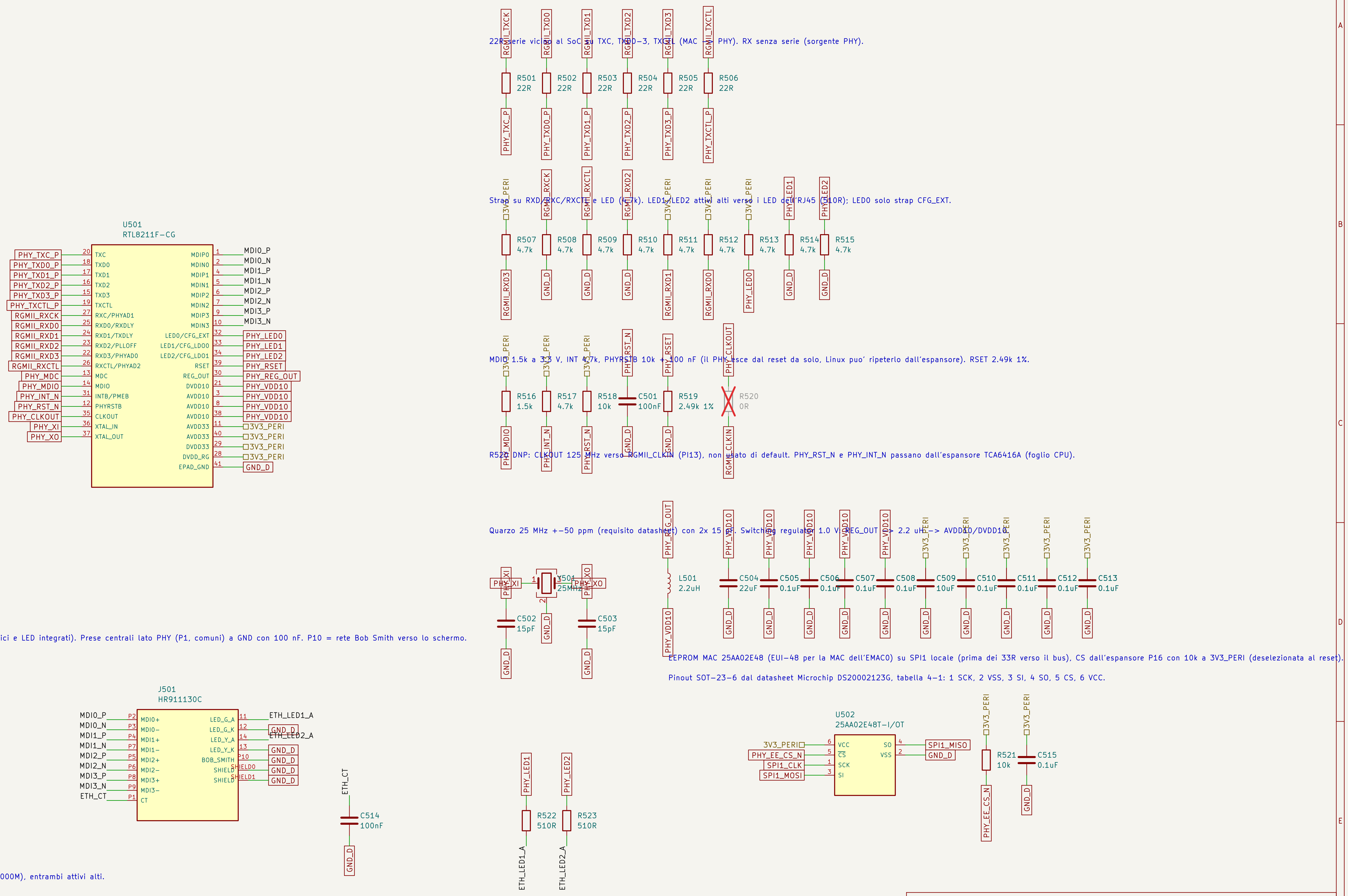
RETE: PHY Realtek RTL8211F-CG (datasheet rev. 1.7) su RGMII EMAC0 banco PI (VCC_PI 3.3 V). Serie 5xx. Net RGMII_*/PHY_* globali verso il foglio CPU.

Strap (4.7k): PHYAD = 001, PLLOFF = 0, TXDLY = 1 e RXDLY = 1 (rgmii-id, ritardi nel PHY), CFG_EXT = 1 con CFG_LDO = 00: pad RGMII alimentati da fuori a 3.3 V.


U501
RTL8211F-CG
PHY_TXC_P 20 TXC
PHY_TXD0_P 18 TXD0
PHY_TXD1_P 17 TXD1
PHY_TXD2_P 16 TXD2
PHY_TXD3_P 15 TXD3
PHY_TXCTL_P 19 TXCTL
RGMII_RXCK 27 RXC/PHYAD1
RGMII_RXD0 25 RXD0/RXDLY
RGMII_RXD1 24 RXD1/TXDLY
RGMII_RXD2 23 RXD2/PLLOFF
RGMII_RXD3 22 RXD3/PHYAD0
RGMII_RXCTL 26 RXCTL/PHYAD2
PHY_MDC 13 MDC
PHY_MDIO 14 MDIO
PHY_INT_N 31 INTB/PMEB
PHY_RST_N 12 PHYRSTB
PHY_CLKOUT 35 CLKOUT
PHY_XI 36 XTAL_IN
PHY_XO 37 XTAL_OUT
MDIP0 1 MDI0_P
MDIN0 2 MDI0_N
MDIP1 4 MDI1_P
MDIN1 5 MDI1_N
MDIP2 6 MDI2_P
MDIN2 7 MDI2_N
MDIP3 9 MDI3_P
MDIN3 10 MDI3_N
LED0/CFG_EXT 32 PHY_LED0
LED1/CFG_LDO0 33 PHY_LED1
LED2/CFG_LDO1 34 PHY_LED2
RSET 39 PHY_RSET
REG_OUT 30 PHY_REG_OUT
DVDD10 21 PHY_VDD10
AVDD10 3 PHY_VDD10
AVDD10 8 PHY_VDD10
AVDD10 38 PHY_VDD10
AVDD33 11 3V3_PERI
AVDD33 40 3V3_PERI
DVDD33 29 3V3_PERI
DVDD_RG 28 3V3_PERI
EPAD_GND 41 GND_D
RGMII_TXCK R501 22R PHY_TXC_P
RGMII_TXD0 R502 22R PHY_TXD0_P
RGMII_TXD1 R503 22R PHY_TXD1_P
RGMII_TXD2 R504 22R PHY_TXD2_P
RGMII_TXD3 R505 22R PHY_TXD3_P
RGMII_TXCTL R506 22R PHY_TXCTL_P
3V3_PERI R507 4.7k RGMII_RXD3
RGMII_RXCK R508 4.7k GND_D
RGMII_RXCTL R509 4.7k GND_D
RGMII_RXD2 R510 4.7k GND_D
3V3_PERI R511 4.7k RGMII_RXD1
3V3_PERI R512 4.7k RGMII_RXD0
3V3_PERI R513 4.7k PHY_LED0
PHY_LED1 R514 4.7k GND_D
PHY_LED2 R515 4.7k GND_D
3V3_PERI R516 1.5k PHY_MDIO
3V3_PERI R517 4.7k PHY_INT_N
3V3_PERI R518 10k PHY_RST_N
PHY_RST_N C501 100nF GND_D
PHY_RSET R519 2.49k 1% GND_D
PHY_CLKOUT R520 0R RGMII_CLKIN
PHY_XI Y501 25MHz PHY_XO
1 2 3
C502 15pF GND_D
C503 15pF GND_D
GND_D
PHY_REG_OUT L501 2.2uH PHY_VDD10
PHY_VDD10 C504 22uF GND_D
PHY_VDD10 C505 0.1uF GND_D
PHY_VDD10 C506 0.1uF GND_D
PHY_VDD10 C507 0.1uF GND_D
PHY_VDD10 C508 0.1uF GND_D
3V3_PERI C509 10uF GND_D
3V3_PERI C510 0.1uF GND_D
3V3_PERI C511 0.1uF GND_D
3V3_PERI C512 0.1uF GND_D
3V3_PERI C513 0.1uF GND_D
J501
HR911130C
MDI0_P P2 MDI0+
MDI0_N P3 MDI0−
MDI1_P P4 MDI1+
MDI1_N P7 MDI1−
MDI2_P P5 MDI2+
MDI2_N P6 MDI2−
MDI3_P P8 MDI3+
MDI3_N P9 MDI3−
ETH_CT P1 CT
LED_G_A 11 ETH_LED1_A
LED_G_K 12 GND_D
LED_Y_A 14 ETH_LED2_A
LED_Y_K 13 GND_D
BOB_SMITH P10 GND_D
SHIELD SHIELD0 GND_D
SHIELD SHIELD1 GND_D
ETH_CT C514 100nF GND_D
PHY_LED1 R522 510R ETH_LED1_A
PHY_LED2 R523 510R ETH_LED2_A
U502
25AA02E48T-I/OT
3V3_PERI 6 VCC
PHY_EE_CS_N 5 CS
SPI1_CLK 1 SCK
SPI1_MOSI 3 SI
SO 4 SPI1_MISO
VSS 2 GND_D
3V3_PERI R521 10k PHY_EE_CS_N
3V3_PERI C515 0.1uF GND_D

22R serie vicino al SoC su TXC, TXD0-3, TXCTL (MAC -> PHY). RX senza serie (sorgente PHY).

Strap su RXD/RXC/RXCTL e LED (4.7k). LED1/LED2 attivi alti verso i LED dell'RJ45 (510R); LED0 solo strap CFG_EXT.

MDIO 1.5k a 3.3 V, INT 4.7k, PHYRSTB 10k + 100 nF (il PHY esce dal reset da solo, Linux puo' ripeterlo dall'espansore). RSET 2.49k 1%.

R520 DNP: CLKOUT 125 MHz verso RGMII_CLKIN (PI13), non usato di default. PHY_RST_N e PHY_INT_N passano dall'espansore TCA6416A (foglio CPU).

Quarzo 25 MHz +-50 ppm (requisito datasheet) con 2x 15 pF. Switching regulator 1.0 V: REG_OUT -> 2.2 uH -> AVDD10/DVDD10.

RJ45 HanRun HR911130C (100/1000Base-T, magnetici e LED integrati). Prese centrali lato PHY (P1, comuni) a GND con 100 nF. P10 = rete Bob Smith verso lo schermo.

EEPROM MAC 25AA02E48 (EUI-48 per la MAC dell'EMAC0) su SPI1 locale (prima dei 33R verso il bus), CS dall'espansore P16 con 10k a 3V3_PERI (deselezionata al reset).

Pinout SOT-23-6 dal datasheet Microchip DS20002123G, tabella 4-1: 1 SCK, 2 VSS, 3 SI, 4 SO, 5 CS, 6 VCC.

LED verde = LED1 del PHY (100M), giallo = LED2 (1000M), entrambi attivi alti.

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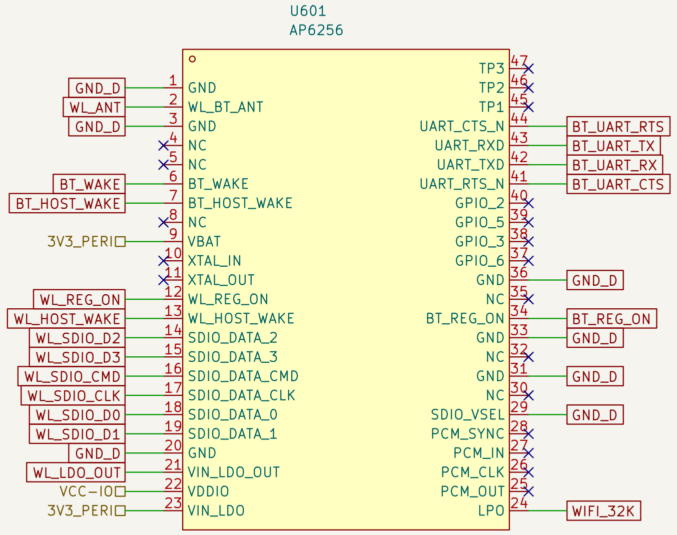
RADIO: AMPAK AP6256 (Wi-Fi 5 + BT 5.0) su SMHC1 PG0-PG5 (SDIO), UART1 PG6-PG9 con RTS/CTS, REG_ON e wake sull'espansore TCA6416A. Serie 6xx. Net WL_*/BT_*/WIFI_32K globali.

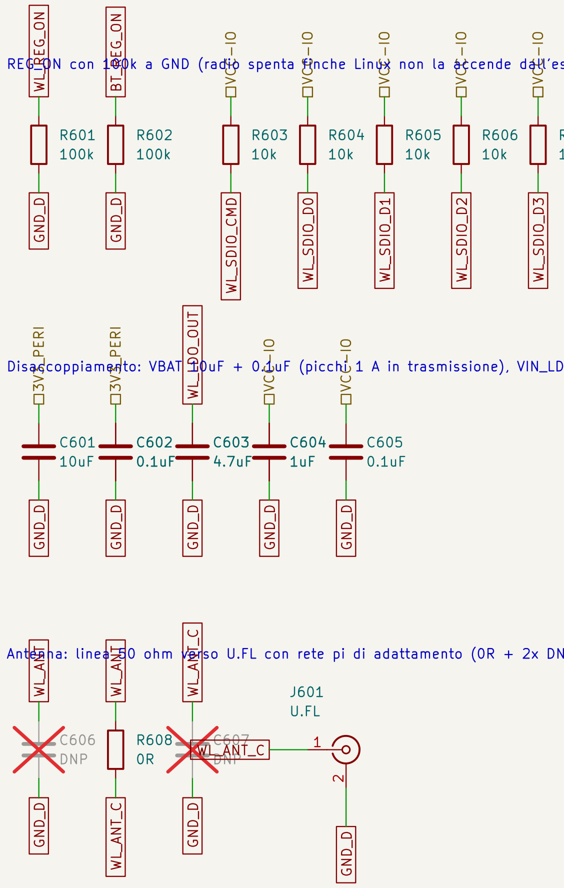
VDDIO = VCC-IO 3.3 V (= VCC_PG), SDIO 2.0 fino a 50 MHz. VBAT e VIN_LDO da 3V3_PERI. LPO 32.768 kHz da X32KFOUT (PG10), derivato dal 24 MHz: da verificare.


U601
AP6256
GND_D 1 GND
WL_ANT 2 WL_BT_ANT
GND_D 3 GND
4 NC
5 NC
BT_WAKE 6 BT_WAKE
BT_HOST_WAKE 7 BT_HOST_WAKE
8 NC
3V3_PERI 9 VBAT
10 XTAL_IN
11 XTAL_OUT
WL_REG_ON 12 WL_REG_ON
WL_HOST_WAKE 13 WL_HOST_WAKE
WL_SDIO_D2 14 SDIO_DATA_2
WL_SDIO_D3 15 SDIO_DATA_3
WL_SDIO_CMD 16 SDIO_DATA_CMD
WL_SDIO_CLK 17 SDIO_DATA_CLK
WL_SDIO_D0 18 SDIO_DATA_0
WL_SDIO_D1 19 SDIO_DATA_1
GND_D 20 GND
WL_LDO_OUT 21 VIN_LDO_OUT
VCC-IO 22 VDDIO
3V3_PERI 23 VIN_LDO
TP3 47
TP2 46
TP1 45
UART_CTS_N 44 BT_UART_RTS
UART_RXD 43 BT_UART_TX
UART_TXD 42 BT_UART_RX
UART_RTS_N 41 BT_UART_CTS
GPIO_2 40
GPIO_5 39
GPIO_3 38
GPIO_6 37
GND 36 GND_D
NC 35
BT_REG_ON 34 BT_REG_ON
GND 33 GND_D
NC 32
GND 31 GND_D
NC 30
SDIO_VSEL 29 GND_D
PCM_SYNC 28
PCM_IN 27
PCM_CLK 26
PCM_OUT 25
LPO 24 WIFI_32K

REG_ON con 100k a GND (radio spenta finche Linux non la accende dall'espansore). Pull-up 10k su CMD e DAT0-3 a VCC-IO.

Disaccoppiamento: VBAT 10uF + 0.1uF (picchi 1 A in trasmissione), VIN_LDO_OUT 4.7uF, VDDIO 1uF + 0.1uF.

Antenna: linea 50 ohm verso U.FL con rete pi di adattamento (0R + 2x DNP) da tarare.


WL_REG_ON
BT_REG_ON
VCC-IO
VCC-IO
VCC-IO
VCC-IO
VCC-IO
R601 100k
R602 100k
R603 10k
R604 10k
R605 10k
R606 10k
R607 10k
GND_D
GND_D
WL_SDIO_CMD
WL_SDIO_D0
WL_SDIO_D1
WL_SDIO_D2
WL_SDIO_D3
3V3_PERI
3V3_PERI
WL_LDO_OUT
VCC-IO
VCC-IO
C601 10uF
C602 0.1uF
C603 4.7uF
C604 1uF
C605 0.1uF
GND_D
GND_D
GND_D
GND_D
GND_D
WL_ANT
WL_ANT
WL_ANT_C
C606 DNP
R608 0R
C607 DNP
WL_ANT_C
J601
U.FL
1
2
GND_D
WL_ANT_C
GND_D
GND_D

UART incrociata: TXD del modulo -> BT_UART_RX (PG7), RXD <- BT_UART_TX (PG6), RTS_N -> BT_UART_CTS (PG9), CTS_N <- BT_UART_RTS (PG8).

PCM (25-28) non collegati: niente audio Bluetooth. XTAL_IN/OUT liberi (quarzo interno). GPIO e TP liberi.

Sheet: /Radio/
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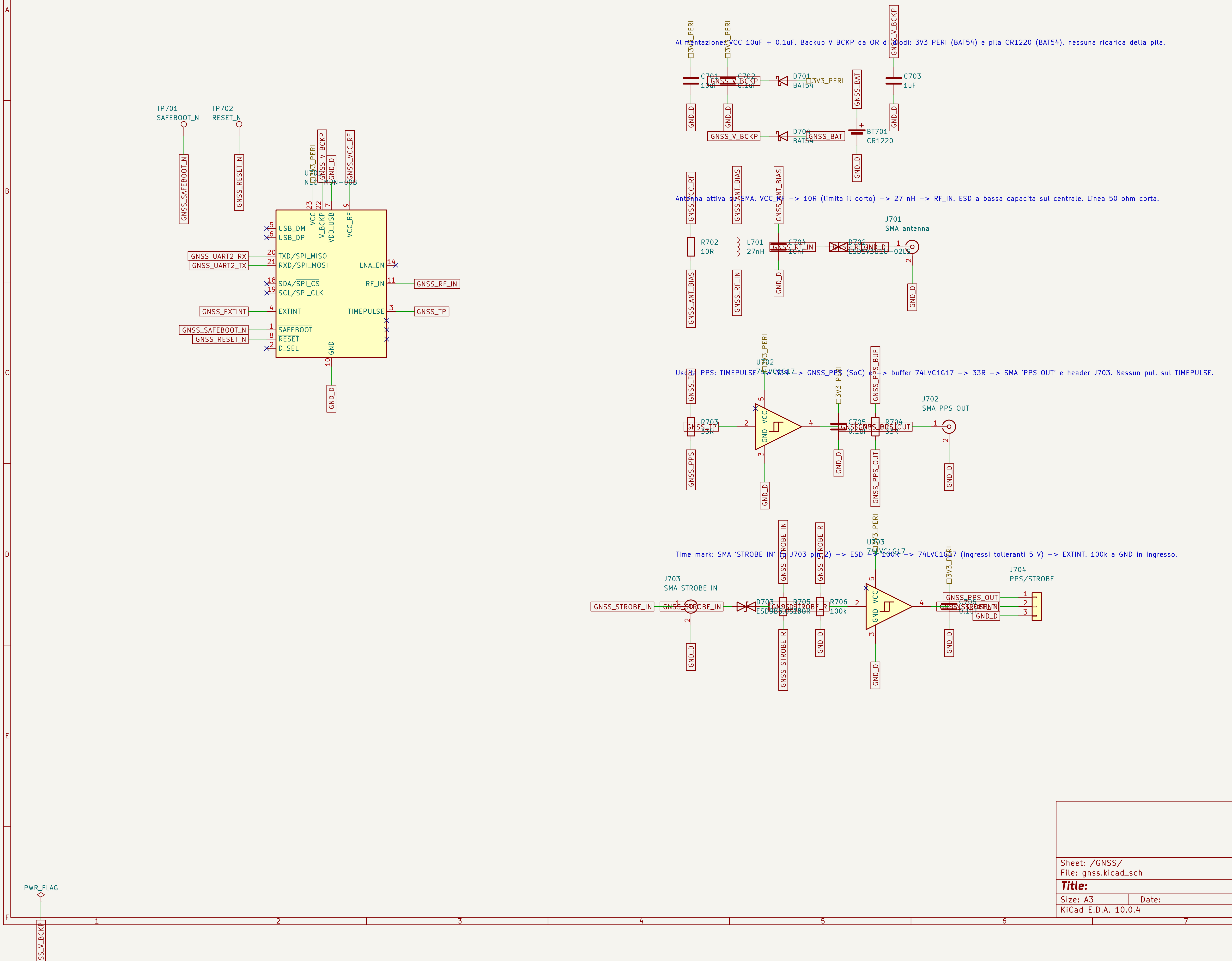
GNSS: u-blox NEO-M9N-00B (formato NEO, stesso footprint del NEO-F10T). Pin dal datasheet UBX-19014285 R08, tabella 9 (= simbolo NEO-M9N di KiCad). Serie 7xx.

UART2: TXD del modulo -> GNSS_UART2_RX (PG16), RXD <- GNSS_UART2_TX (PG15). TIMEPULSE -> 33R -> GNSS_PPS (PG13, EINT). EXTINT (pin 4, PIO7, time mark TIM-TM2) letto anche da PG14 (EINT, solo ingresso).

D_SEL (pin 2) aperto = UART + I2C. V_USB (pin 7) a GND: USB non usata (da confermare sull'integration manual). LNA_EN e riservati liberi.



Alimentazione: VCC 10uF + 0.1uF. Backup V_BCKP da OR di diodi: 3V3_PERI (BAT54) e pila CR1220 (BAT54), nessuna ricarica della pila.

Antenna attiva su SMA: VCC_RF -> 10R (limita il corto) -> 27 nH -> RF_IN. ESD a bassa capacita sul centrale. Linea 50 ohm corta.

Uscita PPS: TIMEPULSE -> 33R -> GNSS_PPS (SoC) e -> buffer 74LVC1G17 -> 33R -> SMA 'PPS OUT' e header J703. Nessun pull sul TIMEPULSE.

Time mark: SMA 'STROBE IN' (o J703 pin 2) -> ESD -> 100R -> 74LVC1G17 (ingressi tolleranti 5 V) -> EXTINT. 100k a GND in ingresso.

TP701
SAFEBOOT_N
TP702
RESET_N
GNSS_SAFEBOOT_N
GNSS_RESET_N
U701
NEO-M9N-00B
3V3_PERI
GNSS_V_BCKP
GND_D
GNSS_VCC_RF
VCC
V_BCKP
VDD_USB
VCC_RF
USB_DM
USB_DP
TXD/SPI_MISO
RXD/SPI_MOSI
SDA/SPI_CS
SCL/SPI_CLK
EXTINT
SAFEBOOT
RESET
D_SEL
GND
LNA_EN
RF_IN
TIMEPULSE
GNSS_UART2_RX
GNSS_UART2_TX
GNSS_EXTINT
GNSS_RF_IN
GNSS_TP
C701
10uF
C702
0.1uF
D701
BAT54
C703
1uF
GNSS_BAT
D704
BAT54
BT701
CR1220
R702
10R
L701
27nH
C704
10nF
D702
ESD5V3U1U-02LS
J701
SMA antenna
GNSS_ANT_BIAS
U702
74LVC1G17
R703
33R
C705
0.1uF
R704
33R
GNSS_PPS_BUF
J702
SMA PPS OUT
GNSS_PPS
GNSS_PPS_OUT
J703
SMA STROBE IN
GNSS_STROBE_IN
D703
ESD9B5.0ST5G
R705
100R
GNSS_STROBE_R
R706
100k
U703
74LVC1G17
C706
0.1uF
J704
PPS/STROBE
PWR_FLAG

Sheet: /GNSS/
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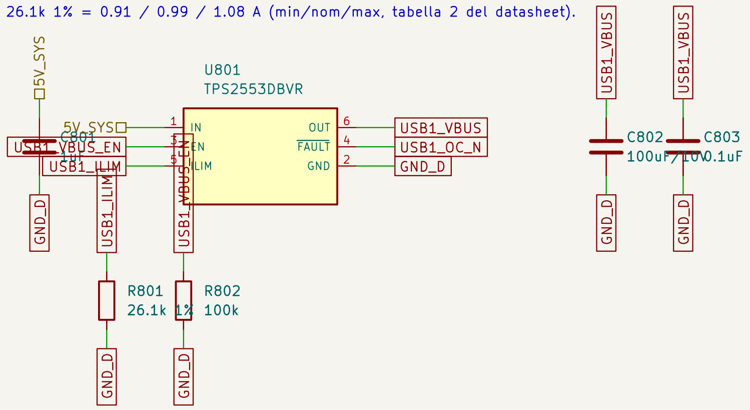
USB HOST: USB1 del SoC su USB−C (16 pin, HRO TYPE−C−31−M−12), solo host, come SchedaA133. Serie 8xx. Net USB1_* globali verso il foglio CPU.

La porta non alimenta mai la scheda: VBUS esce solo dal load switch. Rp fissi da 22k a 5V_SYS su CC1/CC2 = sorgente 1.5 A (Type−C), niente controller PD.

TPS2553: EN attivo alto dall’espansore (P03) con 100k a GND, FAULT open−drain su PC12 (pull−up 10k a VCC−1V8 sul foglio CPU).

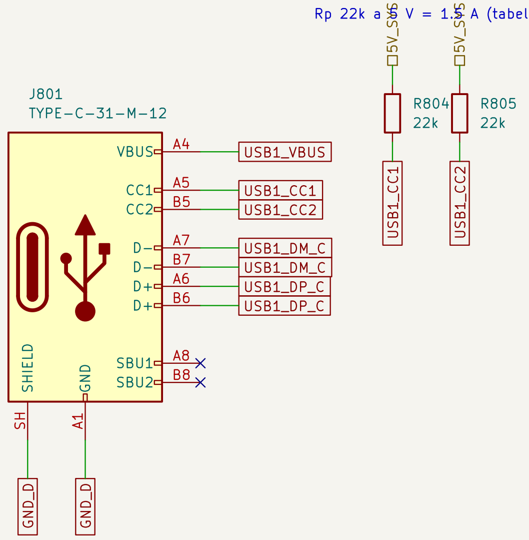
RILIM 26.1k 1% = 0.91 / 0.99 / 1.08 A (min/nom/max, tabella 2 del datasheet).



5V_SYS
U801
TPS2553DBVR
5V_SYS
C801
1uF
USB1_VBUS_EN
USB1_ILIM
1
3
5
IN
EN
ILIM
OUT
FAULT
GND
6
4
2
USB1_VBUS
USB1_OC_N
GND_D
GND_D
USB1_ILIM
USB1_VBUS_EN
R801
26.1k 1%
R802
100k
GND_D
GND_D
USB1_VBUS
USB1_VBUS
C802
100uF/10V
C803
0.1uF
GND_D
GND_D

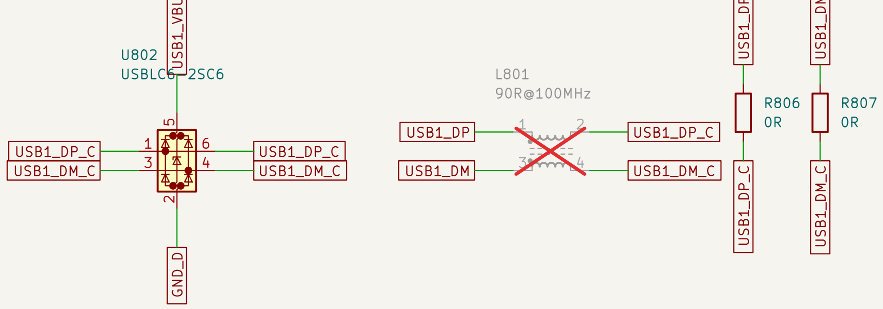
J801 USB−C: DP/DM collegati su entrambi i lati (A6/B6, A7/B7), SBU non connessi, schermo a GND.

Rp 22k a 5 V = 1.5 A (tabella Type−C). Il TPS2553 limita comunque a circa 1 A.



5V_SYS
5V_SYS
R804
22k
R805
22k
USB1_CC1
USB1_CC2
J801
TYPE−C−31−M−12
VBUS
A4
USB1_VBUS
CC1
A5
USB1_CC1
CC2
B5
USB1_CC2
D−
A7
USB1_DM_C
D−
B7
USB1_DM_C
D+
A6
USB1_DP_C
D+
B6
USB1_DP_C
SBU1
A8
SBU2
B8
SHIELD
GND
SH
A1
GND_D
GND_D

ESD USBLC6−2SC6 vicino al connettore su DP/DM e VBUS. Choke di modo comune DNP con 0R in parallelo (footprint sovrapposti).



USB1_VBUS
U802
USBLC6−2SC6
5
1
3
2
6
4
USB1_DP_C
USB1_DM_C
USB1_DP_C
USB1_DM_C
GND_D
L801
90R@100MHz
USB1_DP
USB1_DM
1
2
3
4
USB1_DP_C
USB1_DM_C
USB1_DP
USB1_DM
R806
0R
R807
0R
USB1_DP_C
USB1_DM_C

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